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Education

01/2020 to 11/2024 **PhD in Theoretical Astroparticle Physics with distinction (summa cum laude)**
DESY Zeuthen
Within the [Helmholtz Weizmann Research School on Multimessenger Astronomy](#)
Supervisors: Walter Winter, Andrew Taylor, Eli Waxman
Focus: *Time-Dependent Radiation Modelling of Gamma-Ray Burst Afterglows*

- Reduced analytical afterglow modelling
- Time-dependent afterglow modelling of gamma-ray bursts
- Development, maintenance and publication of the [AM³ code](#)
- Counts-level fits from keV-TeV energies

10/2016 to 12/2019 **Master of Science in Physics with distinction**
Focus: Astroparticle Physics and Cosmology
RWTH Aachen University
Supervisors: Thomas Bretz, Philipp Mertsch
Thesis: *Developing a Magnetic Reconnection Model to Interpret Blazar Variability as Measured with the FACT Telescope*

10/2013 to 10/2016 **Bachelor of Science in Physics with distinction**
RWTH Aachen University
Supervisor: Christopher Wiebusch
Thesis: *Development of a measurement setup to compare acoustic emitters for the EnEx-RANGE project*

Professional Experience

10/2024 to present **Postdoctoral Researcher**
Anton Pannekoek Institute for Astronomy / University of Amsterdam
Advisors: Sera Markoff
Focus: *Improving multi-messenger physics in GRMHD simulations of black holes*

06/2022 **Summer School - Foundations of Cosmic Ray Astrophysics, Varenna (Italy)**

05/2017 - 02/2019 **Assistant Scientist at EnEx RANGE Project at RWTH Aachen**
Focus: Experimental lab work and data analysis
Glacier expedition for project validation (08/2017)

02/2017 - 03/2017 **Site visit of Pierre-Auger-Observatory, Argentina**
Participation in maintenance, shifts, collaboration meeting and public outreach work at the local visitor's facility

09/2016 - 12/2016 **Scientific stay at the HAWC Observatory, Mexico**
Studies on Radio Galaxies at the National Autonomous University of Mexico (UNAM)
Participation in the *HAWC Collaboration Meeting* and the *Workshop on a wide field-of-view Southern Hemisphere TeV gamma ray observatory*

Awards

12/2020	Financial Performance Bonus for Active and Excellent Engagement in Research Discussions - Deutsches Elektronen-Synchrotron (DESY)
2013	DPG-Abiturpreis Physics - High school award of German Physical Society

Teaching Experience

07/2023 - 09/2023	DESY Summer student program
07/2022 - 09/2022	Supervision of 3 summer student projects on GRB afterglow modelling
10/2020 - 02/2021	Mathematical Tools for a Physicist at Humboldt University Preparation and discussion of weekly exercise sheets

Outreach

2022, 2021	International Cosmic Day (yearly event) Talk and supervision of a cosmic-ray experiment in German high school
02/2022	Attended Workshop on Outreach Skills 3-day workshop of the German network of (Astro)particle physics outreach
2018 - 2020	AIS³ - Astroparticle Immersive Synthesizer³ Construction and Maintenance of AIS³ with Tim Otto Roth in Berlin (08-09/2018), Munich (02/2019) and Aachen (fall 2019) including outreach talks for school classes

Skills

Computer Skills	Python, C++ LaTeX, Microsoft Office (Excel, Word, PowerPoint), Markdown, GIT	
Languages	German (<i>mother tongue</i>) English (<i>fluent</i>)	Spanish (<i>advanced</i>) Dutch (<i>basic</i>)

Intercultural Experience

08/2019	Attended Workshop on intercultural communication (2 days) Combining theoretical concepts with practical application to train the ability to detect and resolve intercultural communicational problems
11/2018	German Italian Physics Exchange (GIPE) to Trieste, Italy (4 days)
10/2018 - 03/2019	Mentoring of international master students in Aachen
04/2016 - 09/2016	Support of a Spanish exchange student in Aachen (BeBuddy)
11/2014	Intercultural training (2 days)
06/2012 - 08/2012	School exchange to Instituto de Ballester, Buenos Aires, Argentina

Talks

16/10/2024	NOVA Network 3 meeting, Nikhef Amsterdam Lepto-hadronic modelling of gamma-ray burst afterglows
14/10/2024	GRAPPA journal club, GRAPPA Amsterdam The Astrophysical Multi-Messenger Modelling (AM ³) Software
21/02/2024	Invited: Workshop on Numerical Multi-messenger Modeling, Paris News from AM ³
28/11/2023	Invited: SPIMAX Seminar, Oxford On the camel nature of GRB afterglows
24/10/2023	Plenary: High Energy Phenomena in Relativistic Outflows (HEPRO) Lepto-Hadronic Very-High-Energy GRB Afterglows
30/06/2023	Gamma Group Seminar, DESY Zeuthen GRB 221009A: Afterglow spectrum
21/02/2023	Invited: DESY Astroparticle division retreat, Potsdam, Germany Intro-slides for critical review discussion on transients (with R. Konno)
26/01/2023	Science Club of Multimessenger School (online, incl. Weizmann Institute, Israel) The Weizsäcker-Williams Approach
10/01/2023	Invited: Transient Tuesday, Niels Bohr Institute, Copenhagen (online) VHE GRB Afterglows: A story about Bactrians, Dromedaries and lots of Butterflies
18/11/2022	Invited: Astroparticle Seminar at DESY, Zeuthen VHE GRB Afterglows: A story about Bactrians, Dromedaries and lots of Butterflies
16/05/2022 18/05/2022	Talks at the Harnack-Haus GRB Workshop Berlin • One Zone Basics and Effective Descriptions • Multi-wavelength Fitting at the Counts Level
27/10/2021	TeVPA 2021 (online due to Covid) Understanding the Spectrum of Gamma-Ray Burst 190114C
23/09/2021	DESY Theory Workshop: Bright ideas for a dark universe (online due to Covid) Understanding the Spectrum of Gamma-Ray Burst 190114C
22/04/2021	Science Club of Multimessenger School (online, incl. Weizmann Institute, Israel) Relativistic Shocks
22/01/2020	Theory Group Seminar, DESY Zeuthen Blazar Variability with Magnetic Reconnection
26/03/2019	German Physical Society Meeting FACT - Spectral Variability of TeV-Blazars

Posters

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| 06/06/2023 | Annual Meeting of the Multimessenger School in Israel
Understanding the Spectrum of Gamma-Ray Burst 190114C |
| 12/05/2022 | 7th Meeting of the Astroparticle Physics Committee at DESY
Understanding the Spectrum of Gamma-Ray Burst 190114C |

List of Publications

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| 2024 | Leptohadronic Scenarios for TeV Extensions of Gamma-Ray Burst Afterglow Spectra
M. Klinger, C. Yuan, A. M. Taylor and W. Winter (ApJ 977 242) |
| 2024 | AM³: An open-source tool for time-dependent lepto-hadronic modelling of astrophysical sources
M. Klinger, A. Rudolph, X. Rodrigues, et al. (software, ApJS 275 4) |
| 2024 | The Multiwavelength Picture of GRB 221009A's Afterglow
M. Klinger, A. M. Taylor, T. Parsotan, et al. (MNRAS 529) |
| 2023 | Probing the multiwavelength emission scenario of GRB 190114C
M. Klinger, D. Tak, A. M. Taylor and S. J. Zhu (MNRAS 520 1) |
| 2020 | FACT - Probing the Periodicity of Mrk 421 and Mrk 501
M. Klinger, M. Beck and F. Theißen on behalf of the FACT collaboration (PoS(ICRC2019)) |
| 2017 | Observation of radio galaxies with HAWC
D. Avila, R. Alfaro, A. Galván, M.M. González, N. Fraija and M. Klinger on behalf of the HAWC Collaboration (PoS(ICRC2017)) |